MLF Week-8 SWU Session Notes

Q.1: If $f(6) = 3$, $f'(6) = 1/2$, & $f''(6) = 2$.

What could be the value of $f(7)$?

Sol^n:- Taylor series —

$$f(x+\underline{\underline{\eta}}d) = f(x) + \eta d\, f'(x) + \frac{\eta^2 d^2}{2} f''(x) + \dots$$

$$f(7) = f(6+1) \simeq f(6) + 1 \cdot f'(6) + f''(6) \cdot \frac{1}{2}$$

$$f(7) \simeq 3 + 1 \cdot \frac{1}{2} + \frac{2}{2} = 4.5$$

$$\boxed{f(7) \simeq 4.5}$$

Q.2 : $f(x,y) = xy^2$, $\eta = 1$, $(x_0, y_0) = (2,2)$

Gradient Descent Algorithm. $x_{t+1} = x_t - \eta f'(x_t)$

$$\underset{\text{vector}}{\underset{\uparrow}{x_{t+1}}} = \underset{\text{vector}}{\underset{\uparrow}{x_t}} - \eta \cdot \underset{\text{vector.}}{\underset{\uparrow}{\nabla f(x_t)}}$$

$$\nabla f(x,y) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} y^2 \\ 2xy \end{bmatrix}$$

$$\nabla f(x,y)_{x_0=2, y_0=2} = \begin{bmatrix} 4 \\ 8 \end{bmatrix} = \nabla f(x_0, y_0) = \nabla f(2,2)$$

First Iteration:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - 1 \cdot \nabla f(x_0, y_0)$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 4 \\ 8 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}$$

Second iteration:

3. $$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} - \eta \nabla f(x_1, y_1) \quad \text{--- } (*)$$

$$\nabla f(x,y) = \begin{bmatrix} y^2 \\ 2xy \end{bmatrix} \Rightarrow \nabla f(x_1, y_1) = \begin{bmatrix} 36 \\ 24 \end{bmatrix}$$

From $(*)$,

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \end{bmatrix} - 1 \cdot \begin{bmatrix} 36 \\ 24 \end{bmatrix} = \begin{bmatrix} -38 \\ -30 \end{bmatrix}$$

$$f(x_2, y_2) = x_2 \cdot y_2^2 = (-38) \cdot (-30)^2$$

$$= -34200.$$

Q.3: $f(x,y) = 2xy + 2x - x^2 - 2y^2$, $\eta = 0.2$

$$(x_0, y_0) = (-1, -1)$$

$$\nabla f(x,y) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2y + 2 - 2x \\ 2x - 4y \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - \eta \cdot \nabla f(x_0, y_0)$$

$$= \begin{bmatrix} -1 \\ -1 \end{bmatrix} - 0.2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} -1.4 \\ -1.4 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} - \eta \cdot \nabla f(x_1, y_1)$$

$$= \begin{bmatrix} -1.4 \\ -1.4 \end{bmatrix} - 0.2 \begin{bmatrix} 2 \\ 2.8 \end{bmatrix}$$

4

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} -1.4 \\ -1.4 \end{bmatrix} - \begin{bmatrix} 0.4 \\ 0.56 \end{bmatrix} = \begin{bmatrix} -1.8 \\ -1.96 \end{bmatrix}$$

$\rightarrow$ $\boxed{x_2 = -1.8}$ Ans.

Q5: $x_0 = 2, \quad \eta = 0.05$

$$f(x) = (2x-8)^2 = 4x^2 + 64 - 32x$$

$\underbrace{\text{error}} = \underbrace{x_k - 4}$ $\boxed{|\text{error}| \le 0.1}$

$$f'(x) = 8x - 32 = 8(x-4)$$

$$x_{k+1} = x_k - \eta \cdot f'(x_k)$$

$$x_{k+1} = x_k - 0.05 \; 8(x_k - 4)$$

$$x_1 = x_0 - 0.05 \cdot 8(x_0 - 4)$$

$$= 2 - 0.4(2-4) = 2 - 0.4(-2) = 2.8$$

$$|2.8 - 4| = 1.2$$

$x_2 = x_1 - 0.4(x_1 - 4) = 2.8 - 0.4(2.8-4)$

6.

$$\boxed{x_2 = 3.28}$$

$$|3.28 - 4| = 0.72$$

$x_3 = 3.28 - 0.4(3.28 - 4) = 3.568$

$$|3.568 - 4| = 0.432$$

$x_4 = 3.7408$

$$|3.7408 - 4| = 0.25$$

$x_5 = 3.8448$

$$|3.8448 - 4| = 0.156$$

$x_6 = 3.90666$

$$|3.90666 - 4| < \underline{\underline{0.1}}$$

After 6 iterations, we reach $x_6 \simeq 3.907$, which is within 0.1 of the $x^* = 4$.

Q.6 :- $\underbrace{f(6)}=3$, $\underbrace{f'(6)}=\frac{1}{2}$, $\underbrace{f''(6)}=2$

7. $x_0 = 6$

$$x^* = 6 - \frac{f'(6)}{f''(6)} = 6 - \frac{1/2}{2} = 6 - \frac{1}{4}$$

$$= 6 - 0.25$$

$$\boxed{x_1 = 5.75}$$

f is quadratic with constant second derivative,

$\Rightarrow$ Newton's method reaches the exact minimizer in one iteration.

$$\boxed{x_1 = 5.75}$$ at first step. 5.75

$$\underbrace{\left| x_1 - 5.75 \right|} = 0$$

Q. $f(x) = x^3 - x^2 - x + 5$

8.

$$f'(x) = 3x^2 - 2x - 1$$

$$f'(x) = 3x^2 - 2x - 1 = 0$$

$$\Rightarrow 3x^2 - 3x + x - 1 = 0$$

$$\Rightarrow 3x(x-1) + 1(x-1) = 0$$

$$\Rightarrow (3x+1)(x-1) = 0$$

$x = 1$ or $x = -1/3$

$$f''(x) = 6x - 2$$

$f''(1) = 6 - 2 = 4 > 0 \Rightarrow$ minimum.

$\rightarrow$ $x_0 = 0.75$, $\eta = 0.25$,

$f'(x) = 3x^2 - 2x - 1$ $\qquad |x_t - 1| \leq 0.001$.

$x_{t+1} = x_t - \eta f'(x_t)$

$x_1 = x_0 - \eta f'(x_0)$

$= 0.75 - 0.25\left(3(0.75)^2 - 2(0.75) - 1\right)$

$\boxed{x_1 = 0.953125}$

error: $|x_1 - 1| = 0.04$

$\rightarrow$ $x_2 = 0.953125 - 0.25\left[3(0.953125)^2 - 2(0.953125) - 1\right]$

$x_2 = 0.998352$

error: $|x_2 - 1| \leq 0.001$ check?

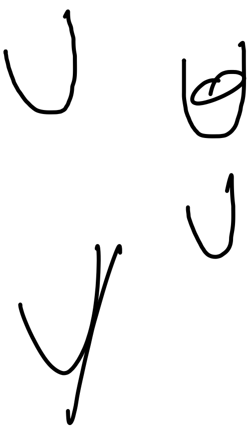
$\rightarrow$ $x_3 =$

error: $|x_3 - 1| \leq 0.001$ or not

General:

$$f(x) = ax^2 + bx + c$$

10.



$$f'(x) = 2ax + b \rightarrow \text{slope}$$

$$\boxed{f''(x) = 2a} \rightarrow \boxed{\text{curvature}}$$

$f''(x) = 2ax$

$\cancel{f''(x) =}$

$f''(a) = 2a$

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

$$x_{n+1} = x_n - \frac{2ax_n + b}{2a}$$

$$= x_n - \frac{\cancel{2a}}{\cancel{2a}} \cdot x_n - \frac{b}{2a}$$

$$\boxed{x_{n+1} = -\frac{b}{2a}} \rightarrow$$

$$f = f(x) + f'(x) + \frac{f''(x)}{2} + \frac{f'''(x)}{3!} + \dots$$

$f''(a) = 2a$

$f''(x) = 2ax^2 \Rightarrow f'''(x)$

Newton:

11. $f(x) = x^3 - x^2 - x + 5$

$f'(x) = 3x^2 - 2x - 1$

$f''(x) = \underline{\underline{6x - 2}}$ $\longrightarrow f'''(\underline{x})$

$x_0 = \underline{\underline{0.75}}$ $f'(0.75) = 3(0.75)^2 - 2(0.75) - 1$

$= -0.8125$

$f''(0.75) = 6(0.75) - 2 = 2.5$

$x_1 = 0.75 - \dfrac{-0.8125}{2.5} = 0.75 + \bigcirc = 1.075$

$|x_1 - 1| \not\leq 0.001$

$x_2 =$

$x_3 =$ $\| x_3 - 1 \,| \leq \underline{\underline{0.001}}$ or $\underline{\text{not}}$.

12.

$x_1 = 5.75$. $|x_1 - 5.75| = 0 \leq 0.1$

≤ 0.005

$f''(6) = 2$

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